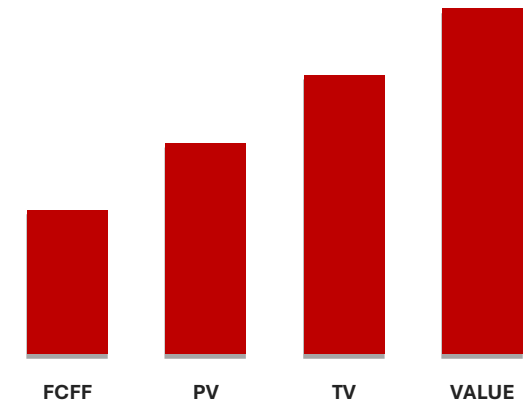


Corporate Valuation Methods

Session 3

Intrinsic Valuation

From forecast cash flows to enterprise value, equity value and value per share



Key Assumptions Behind a DCF Valuation

1. GOING CONCERN

The firm is expected to continue operating rather than liquidate in the near future.

2. FORECASTABLE CASH FLOWS

Revenue, margins, taxes and reinvestment can be estimated using defensible assumptions.

3. RISK IS PRICED

The discount rate reflects the opportunity cost and risk of the cash flows being valued.

4. CONSISTENCY

FCFF → WACC → Enterprise Value
FCFE → Cost of Equity → Equity Value

5. VARYING GROWTH

Growth can change year by year during the explicit forecast period and should move toward maturity.

6. STABLE TERMINAL STATE

A constant sustainable growth rate is required only for the perpetual-growth terminal period.

DCF does not assume one constant growth rate throughout the model.

The DCF Valuation Map

FORECAST

$FCFF_1 \dots FCFF_n$

DISCOUNT

PV of explicit
FCFF

TERMINAL

Value after
Year n

ENTERPRISE

PV (FCFF) +
PV (TV)

EQUITY

EV less
claims +
non-
operating
assets

PER SHARE

Equity ÷
diluted
shares

Value today = Present value of all future cash flows

DCF is simply a present-value framework.

Keep the valuation object clear at every step: firm cash flow produces firm value.

Firm Value vs Equity Value — Do Not Mix the Two

FCFF APPROACH

- Cash flow available to debt and equity investors.
- Discount rate: WACC.
- Output: enterprise / firm value.
- Then bridge from enterprise value to equity value.

FCFE APPROACH

- Cash flow available to common equity holders.
- Discount rate: cost of equity.
- Output: equity value directly.
- No debt subtraction after the valuation.

FCFF $\leftrightarrow$ WACC $\leftrightarrow$ Enterprise Value

FCFE $\leftrightarrow$ Ke $\leftrightarrow$ Equity Value

FCFF Valuation — The Core Equation

$$EV = \sum [FCFF_t / (1 + WACC)^t] + TV_n / (1 + WACC)^n$$

Explicit-period cash flows + value of all cash flows after the explicit period

FCFF_t

Forecast free cash flow to the firm in each explicit year.

WACC

Opportunity cost of capital for the operating assets.

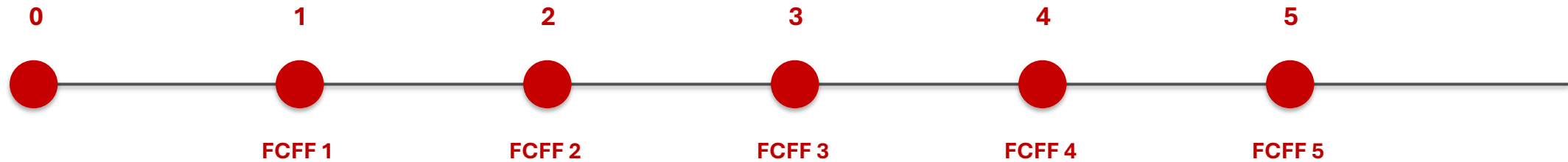
TV_n

Value at the end of Year n of all cash flows from Year n+1 onward.

The terminal value is measured at Year n — it still has to be discounted back to today.

A Timeline Makes the Discounting Clear

Valuation date



TV₅

TV₅ contains cash flows
from Year 6 onward.

Year 5 FCFF is an explicit cash flow. Terminal value starts with Year 6 cash flow.

Worked Step 1 — Discount the Explicit FCFF

$$PV(FCFF_t) = FCFF_t / (1 + WACC)^t$$

Year	1	2	3	4	5
FCFF (€m)	90	100	112	124	135
Discount factor @ 8%	0.926	0.857	0.794	0.735	0.681
PV of FCFF (€m)	83.3	85.7	88.9	91.1	91.9

PV of explicit FCFF = €440.9m. We still need the value of cash flows after Year 5.

Why Do We Need a Terminal Value?

THE PROBLEM

A going concern may generate cash flows long after the detailed forecast ends.

Stopping the model at Year 5 would treat all later cash flows as zero.

THE SOLUTION

Forecast in detail only while assumptions can be defended.

At the end of that period, move the company toward a stable state.

THE TERMINAL VALUE

Compress all post-forecast cash flows into one value at the end of the explicit period.

Terminal value is continuation value — not an arbitrary plug and not necessarily liquidation value.

When Should the Explicit Forecast End?

NOT AUTOMATICALLY YEAR 5

- Five years is a convention, not an economic law.
- Use a longer period if the company is still far from a stable operating state.

LOOK FOR TRANSITION

- Growth moving toward sustainable levels.
- Margins approaching a defensible mature level.
- Reinvestment consistent with long-run growth.
- Risk and financing becoming more stable.

ASK ONE QUESTION

At the end of the explicit period, can these assumptions reasonably persist for a long time?

The forecast horizon should be long enough to reach a credible mature state.

Stable Growth — What Must Become Sustainable?

The explicit forecast may use varying growth rates; these assumptions apply only once the firm reaches the terminal stable state.

GROWTH

A long-run rate compatible with the currency and economy.

MARGIN

A mature operating margin, not a temporary peak.

ROIC

A sustainable return on invested capital.

REINVESTMENT

Enough investment to support the chosen growth rate.

RISK / WACC

Risk characteristics consistent with a mature firm.

Stable growth is a package of assumptions — not just one number called g .

Perpetual-Growth Terminal Value

$$TV_n = FCFF_{n+1} / (WACC_{stable} - g)$$

Require $WACC > g$

$FCFF_{n+1}$

The first cash flow in the stable period — one year after the terminal year.

g

Sustainable perpetual growth rate, stated consistently with the valuation currency.

$WACC_{stable}$

Cost of capital appropriate for the stable business.

Do not insert Year n FCFF into a formula that requires Year n+1 FCFF.

Growth Requires Reinvestment — Even in Terminal Value

$$\text{Growth} = \text{Reinvestment Rate} \times \text{ROIC}$$

$$\text{Stable Reinvestment Rate} = g / \text{Stable ROIC}$$

For FCFF, reinvestment is measured relative to NOPAT.

EXAMPLE

If $g = 3\%$ and stable ROIC = 10%:
Reinvestment rate = $3\% \div 10\% = 30\%$

INTERPRETATION

The firm must reinvest 30% of NOPAT to sustain 3% growth if new investment earns 10%.

Growth is not free: higher sustainable growth normally requires more reinvestment.

Worked Step 2 — Build the First Stable-Year FCFF

ASSUMPTIONS

Year 5 NOPAT = €150m
Stable growth $g = 3\%$
Stable ROIC = 10%
Stable WACC = 8%

REINVESTMENT

Year 6 NOPAT = 150×1.03
= €154.5m

Reinvestment rate = $3\% \div 10\%$
= 30%

Reinvestment = $154.5 \times 30\%$
= €46.35m

FCFF₆

FCFF = NOPAT – Reinvestment

= $154.5 - 46.35$

= €108.15m

Build FCFF_{n+1} from stable economics before applying the perpetuity formula.

Worked Step 3 — Calculate and Discount Terminal Value

$$TV_5 = €108.15m / (8\% - 3\%) = €2,163m$$

$$PV(TV_5) = €2,163m / (1.08)^5 = €1,472m$$

AT YEAR 5

€2,163m is the value at the end of Year 5 of all FCFF from Year 6 onward.

AT YEAR 0

€1,472m is what that continuation value is worth at the valuation date.

Terminal value must be discounted over the same five periods as the Year 5 cash flow.

Worked Step 4 — Assemble Enterprise Value

PV OF EXPLICIT FCFF

From Years 1–5:

€440.9m

PV OF TERMINAL VALUE

From Year 6 onward:

€1,472.0m

ENTERPRISE VALUE

€440.9m +

€1,472.0m

= €1,912.9m

Enterprise Value = PV of explicit FCFF + PV of terminal value

Terminal Value Is Often Large — That Is Not Automatically an Error

CALCULATE THE SHARE

$PV(TV) \div \text{Enterprise Value}$

Here:

$€1,472 \div €1,912.9$

$\approx 77\%$

WHY IT CAN BE LARGE

A five-year forecast captures only five years of cash flows.

A going concern may operate for decades beyond Year 5.

WHAT TO CHECK

Do not judge TV only by its percentage.

Check the economics behind g, ROIC, reinvestment, margins and WACC.

A large terminal-value share is a reason to inspect assumptions carefully — not automatically reject the model.

Terminal Value — Four Consistency Checks

1 $g < WACC$

The perpetuity denominator must remain positive.

2 REINVESTMENT

Does $g / ROIC$ produce a plausible stable reinvestment rate?

3 MATURE ECONOMICS

Are margins, ROIC and risk sustainable rather than peak-cycle?

4 IMPLIED MULTIPLE

What EV/EBITDA or EV/EBIT multiple is implied by the intrinsic terminal value?

The checks diagnose the assumptions; they do not replace the intrinsic valuation.

Implied Exit Multiple — Use It as a Cross-Check

$$\text{Implied EV / EBITDA} = \text{Terminal Value} \div \text{Terminal-Year EBITDA}$$

START WITH DCF

Calculate terminal value from sustainable FCFF, WACC and g.

TRANSLATE

Divide that intrinsic terminal value by terminal-year EBITDA or EBIT.

INTERPRET

Compare with mature-company economics and relevant market evidence. Investigate large gaps.

Do not choose a market multiple first and call the result intrinsic terminal value.

Enterprise Value Is Not Equity Value

$$\text{Equity Value} = \text{Enterprise Value} - \text{Debt \& Debt-Like Claims} + \text{Cash \& Non-Operating Assets}$$

SUBTRACT

Interest-bearing debt
Other senior claims where relevant
Minority / non-controlling interests where appropriate

ADD

Excess cash
Marketable securities
Other non-operating assets not captured in operating FCFF

BE CONSISTENT

Do not add an asset already generating the FCFF.
Do not subtract a liability already reflected in operating cash flow.

The bridge should reconcile the assets valued in the DCF with the claims on those assets.

Worked Step 5 — Enterprise Value to Equity Value

Bridge item	€m
Enterprise value	1,912.9
Less: debt	(500.0)
Add: excess cash	120.0
Equity value	1,532.9

Equity Value = €1,532.9m

Worked Step 6 — Equity Value to Value per Share

$$\text{Value per Share} = \text{Equity Value} \div \text{Diluted Shares Outstanding}$$

EQUITY VALUE

€1,532.9m

DILUTED SHARES

100m shares

VALUE / SHARE

€15.33

Use a share count consistent with claims that can dilute common shareholders.

FCFE — Value Equity Directly

$$\text{Equity Value} = \sum [\text{FCFE}_t / (1 + K_e)^t] + \text{Terminal Equity Value} / (1 + K_e)^n$$

CASH FLOW

FCFE is the cash flow available to common equity after operating needs, reinvestment and net debt financing.

DISCOUNT RATE

Use the required return on equity, K_e — not WACC.

OUTPUT

The present value is equity value directly. Do not subtract debt again.

Cash flow, discount rate and valuation object must always match.

A DCF Is a Range, Not a Single Precise Number

FORECAST UNCERTAINTY

Revenue, margins and reinvestment can differ from the base case.

DISCOUNT-RATE UNCERTAINTY

WACC is estimated, not observed as a single indisputable number.

TERMINAL UNCERTAINTY

Small changes in stable growth, ROIC and WACC can move terminal value materially.

The model should show how value changes when defensible assumptions change.

Sensitivity Analysis — Change One or Two Key Inputs

Illustrative value/share (€)

g \ WACC	7.0%	7.5%	8.0%	8.5%	9.0%
2.0%	15.9	14.2	12.8	11.6	10.6
2.5%	17.7	15.6	13.9	12.5	11.4
3.0%	19.9	17.3	15.3	13.7	12.3
3.5%	22.8	19.5	17.0	15.1	13.5
4.0%	26.6	22.3	19.2	16.8	14.8

Sensitivity asks: if one or two assumptions move, how does value respond?

Scenario Analysis — Change the Economic Story

DOWNSIDE

- Slower market growth
- Weaker pricing
- Lower margin
- Lower ROIC
- Potentially higher risk

BASE

- Most defensible central operating case
- Assumptions should tell one coherent economic story.

UPSIDE

- Stronger share gains
- Better mix / pricing
- Operating leverage
- Higher ROIC

Scenarios change linked economic drivers; sensitivity changes selected model inputs mechanically.

Dividend Discount Model — Core Valuation Logic

Intrinsic equity value from the present value of expected shareholder distributions.

$$P_0 = \sum [D_t / (1 + k_e)^t]$$

Discount expected dividends at the required return on equity.

VALUATION OBJECT

DDM values common equity directly.

No enterprise-value-to-equity bridge is required.

CASH FLOW

The relevant cash flow is the dividend expected to be paid to common shareholders.

DISCOUNT RATE

Use the cost of equity, k_e , because dividends are equity cash flows.

DDM is a DCF model: the discounted cash flow is the dividend.

Key Assumptions Behind Dividend Discount Models

1. GOING CONCERN

The company is expected to continue operating and distributing value to shareholders.

2. FORECASTABLE DIVIDENDS

Expected future dividends or payout policy can be estimated with reasonable assumptions.

3. EQUITY RISK IS PRICED

The cost of equity appropriately reflects the risk of the dividend stream.

4. PAYOUT IS ECONOMICALLY MEANINGFUL

Dividends should reasonably reflect the firm's sustainable capacity to distribute cash.

5. GROWTH & RETENTION ARE CONSISTENT

Long-run dividend growth must be supported by retained earnings and returns earned on them.

6. STABLE-GROWTH CONDITION

For Gordon growth, the firm must be in a sustainable stable state and k_e must exceed g .

DDM value is only as credible as the dividend, growth and cost-of-equity assumptions.

Three Common Dividend Discount Models

ZERO-GROWTH DDM

Dividend remains constant forever.

$$P_0 = D / k_e$$

Useful mainly as a simple benchmark or for genuinely non-growing distributions.

GORDON GROWTH MODEL

Dividend grows at a constant sustainable rate forever.

$$P_0 = D_1 / (k_e - g)$$

MULTI-STAGE DDM

Dividend growth varies during an explicit period and then converges to stable growth.

Most useful when a firm is transitioning toward maturity.

Choose the model to match the expected pattern of dividend growth.

Gordon Growth Model — Constant Growth DDM

$$P_0 = D_1 / (k_e - g)$$

Requires $k_e > g$ and a sustainable constant growth rate.

D_1 — NEXT DIVIDEND

Use the dividend expected one period from today.

If D_0 is given:
 $D_1 = D_0(1 + g)$

k_e — COST OF EQUITY

Required return for common shareholders.

It must match the risk of the dividend stream.

g — STABLE GROWTH

A perpetual mature-state growth rate.

It is not the firm's current high-growth rate.

The Gordon model applies to stable growth, not automatically to the company's current growth rate.

Gordon Growth — Worked Example

INPUTS

Current dividend $D_0 = €2.00$
Expected stable growth $g = 3.0\%$
Cost of equity $k_e = 9.0\%$

STEP 1 — FIND D_1

$$\begin{aligned} D_1 &= D_0(1 + g) \\ &= €2.00 \times 1.03 \\ &= €2.06 \end{aligned}$$

STEP 2 — VALUE EQUITY

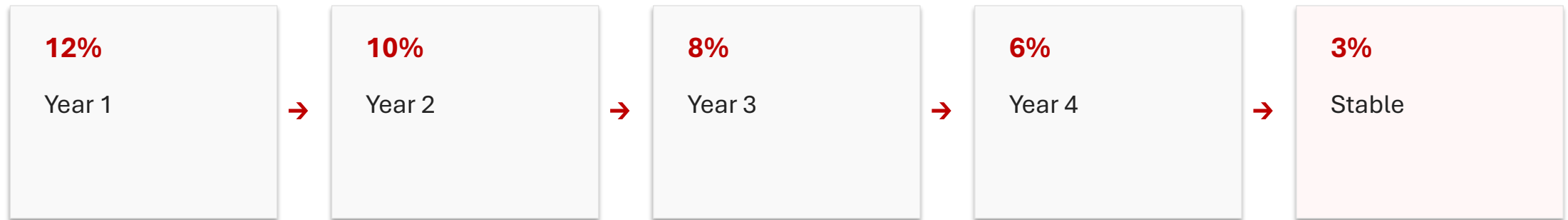
$$\begin{aligned} P_0 &= €2.06 / (0.09 - 0.03) \\ &= €2.06 / 0.06 \\ &= €34.33 \end{aligned}$$

The numerator is D_1 because valuation uses the next expected dividend.

Multi-Stage DDM — Growth Can Vary Before Stability

$$P_0 = \sum_{t=1}^n [D_t / (1 + k_e)^t] + P_n / (1 + k_e)^n$$

Terminal equity value: $P_n = D_{n+1} / (k_e - g_{\text{stable}})$



Forecast changing dividends explicitly; use Gordon growth only after the firm reaches a stable state.

Dividend Growth, Retention and ROE Must Be Consistent

$$g = \text{Retention Ratio} \times \text{ROE}$$

Retention Ratio = $1 - \text{Dividend Payout Ratio}$

EXAMPLE

ROE = 12%

Payout ratio = 60%

Retention ratio = 40%

Sustainable $g = 4.8\%$

WHY RETAIN EARNINGS?

Retained earnings finance equity investment.

Growth depends on both how much is retained and the return earned on it.

CONSISTENCY CHECK

High payout + high perpetual growth requires sufficiently high ROE; otherwise the assumptions conflict.

Growth is not free: dividend payout, retention and ROE jointly determine sustainable growth.

When Is the Dividend Discount Model Appropriate?

MORE APPROPRIATE

- Mature dividend-paying firms
- Relatively stable payout policy
- Dividends linked to sustainable earnings and cash-generation capacity
- Long-run growth can be estimated credibly

LESS APPROPRIATE

- No or very low dividends
- Highly unstable payout
- Dividends disconnected from distributable capacity
- Major changes in leverage or investment policy

ANALYST QUESTION

Are expected dividends a reasonable representation of the value that can ultimately be distributed to common shareholders?

DDM is most informative when payout policy reflects underlying equity economics.